

Project 1 Test Cases

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1 Valid Cases

All Valid cases produce no errors. Both lex and parse complete successfully

1.1 Short and Sweet

```
1 {boolean q
2   print(a)
3
4 }$
```

1.2 Long Test Case

```
1 /*Long Test Case – Everything Except Boolean Declaration */
2 {
3   /* Int Declaration */
4   int a
5   int b
6   a = 0
7   b=0
8   /* While Loop */
9   while (a != 3) {
10    print(a)
11    while (b != 3) {
12      print(b)
13      b = 1 + b
14      if (b == 2) {
15        /* Print Statement */
16        print("there is no spoon" /* This will do nothing */ )
17      }
18    }
19    b = 0
20    a = 1+a
21  }
22 }
23 $
```

Listing 1: Test Case by Alan

2 Invalid Cases

Parsing either fails or program is omitted from CST while other valid programs pass

2.1 Error in Lex

```
1 @@@@Q$
```

Output:

LEX: ERROR Unexpected token '@' at line:1 , column:0
PARSER: ERRORS FOUND IN LEX. UNABLE TO PARSE

2.2 One Program Passes While Another Fails

```
1 {string x ="YELL HEAH" }$  
2 {  
3   int a  
4   a=" hi"  
5 }$
```

Output: Program 0 parse fails. Program 1 parse succeeds. CST is constructed but Program 0 is excluded from tree. (Not copying and pasting :))

2.3 Incomplete Integer Expression

```
1 {  
2   int a  
3   a = 6 +  
4 }$
```

Output: ERROR Expecting [Expr] but received [}] on line:4 column:0

2.4 Initialize and Declaration On Same Line

```
1 {int a= 6}$
```

Output: ERROR Expecting [Statement] but received [=] on line:1 column:6